PRE-IDEATHON TEMPLATE

GENESIS IDEATHON

TEAM & DOMAIN

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- Selected Domain –DOMAIN - 4
- Problem Statement –Smartphone-Based Water Tanker Quality& Pricing Transparency



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DOMAIN PROBLEM IN OUR WORDS

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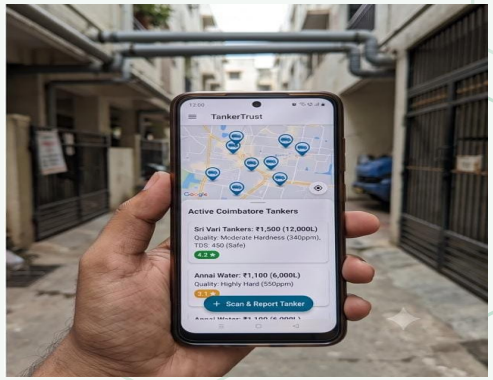
What is needed to be solved?

- Urban households and residents of hostels are not aware of the safety, source and fair pricing of the private water tankers they depend on. They are buying water for higher price even though its not worth for that quality.
- To solve this problem we must equip common consumers with a simple, fail-safe method to instantly check and report tanker water quality (such as hardness and TDS) to prevent abuse.
- We need to give regular people a dead-simple way to test their water quality and log the prices using just their smartphones, so they aren't relying on blind trust.



Why this is important?

- Unsafe, heavily contaminated water is actively hurting thousands of people with serious health damage every day including skin irritation, hair loss and several diseases.
- Today Consumers are being forced to overpay for contaminated water with zero public accountability, and no shared community data to hold suppliers in check.
- Right now, people are forced to overpay for dirty water simply because there’s no community rating system to call out bad suppliers and hold them accountable.



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WHO IS AFFECTED?

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Target users, stakeholders, beneficiaries

Primary users / target group

- Everyday urban families and apartment residents who buy tanker water.
- College hostel wardens and students dealing with daily water issues.
- Small local restaurants and food stalls.



Secondary stakeholders

- Resident Welfare Associations (RWAs) managing the building's maintenance budgets.
- Honest, high-quality private water tanker operators trying to run a fair business.
- Plumbers and maintenance staff who constantly fix calcified pipes.



Who benefits if the problem is solved?

- **The Consumers:** They can stop paying more money for dirty water and avoid severe skin rashes, hair loss, and long-term health issues.
- **The Buildings/Hostels:** They save thousands of rupees by preventing scale buildup and pipe bursts caused by hard water.
- **The Good Suppliers:** Tanker drivers get a verified "safe" rating, bringing them more loyal customers while weeding out the scammers.



CURRENT SITUATION & CONTEXT

Existing scenario, background and how things work today

Existing scenario and background

- During dry months, municipal water supply simply can't keep up with the demands of growing apartments and college hostels.
- To survive, residents are forced to rely on an unorganized, unregulated network of private water tankers.
- Because it's a seller's market, suppliers have total control. They rarely disclose where they pump their water from (often highly saline or polluted borewells) and can change prices on a whim.

Relevant data or facts

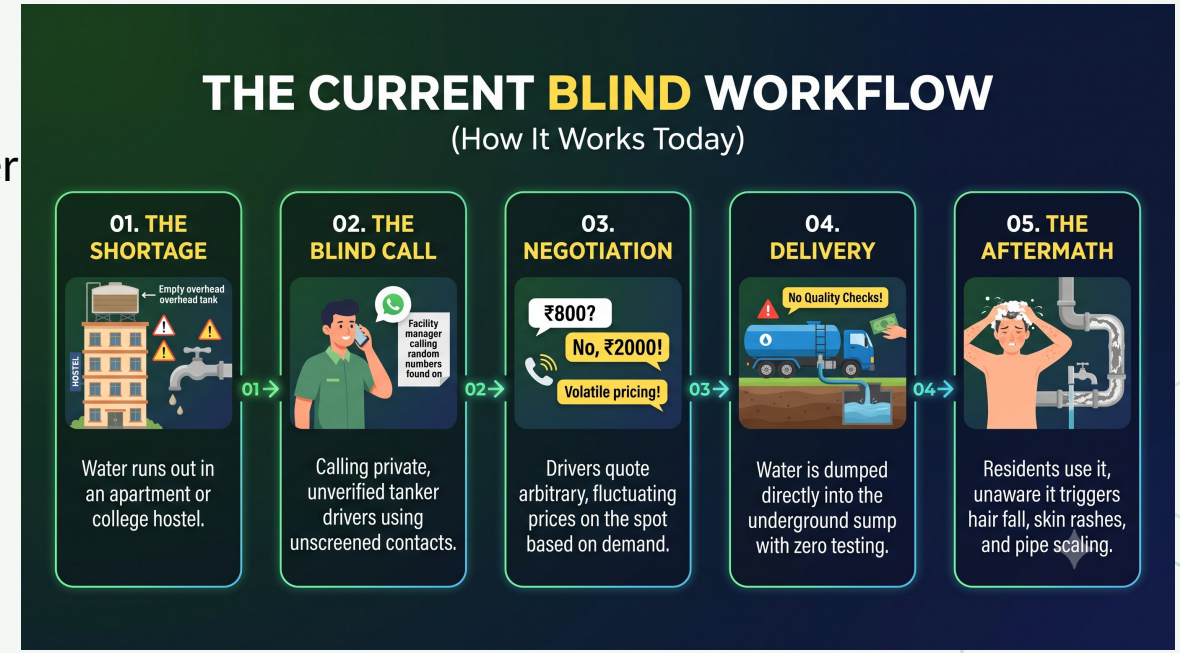
- **Surge Pricing:** The cost of a standard 12,000-liter tanker can swing drastically from ₹800 to over ₹2,000 almost overnight during peak summer.
- **Hidden Health Costs:** Unregulated tanker water often has a TDS (Total Dissolved Solids) level far above the safe limit of 500 mg/L, classifying it as extreme "hard water."
- **Infrastructure Damage:** Continuously using this hard water reduces the lifespan of a building's plumbing and washing machines by up to 30% due to severe calcification.

CURRENT SITUATION & CONTEXT

Existing scenario, background and how things work today

Current workflow, step by step

1. **The Shortage:** An apartment or hostel runs out of their standard water supply.
2. **The Blind Call:** The facility manager calls a private tanker driver (usually just a random number saved from a wall sticker or WhatsApp group).
3. **The Negotiation:** The driver quotes an arbitrary, fluctuating daily price over the phone.
4. **The Unverified Delivery:** The tanker arrives, dumps the water directly into the underground sump, takes the cash, and leaves. Zero quality checks are done.
5. **The Aftermath:** Residents unknowingly use the water, only realizing it's contaminated when their hair starts falling out, their skin breaks out, or their bathroom taps get choked with white scale.



KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

Main difficulties and bottlenecks

- Regular people don't have chemistry labs; traditional water lab tests take days and are way too expensive for daily deliveries.
- There is absolutely no central system or app to compare daily tanker prices or check a supplier’s track record before booking them.

Root causes behind them

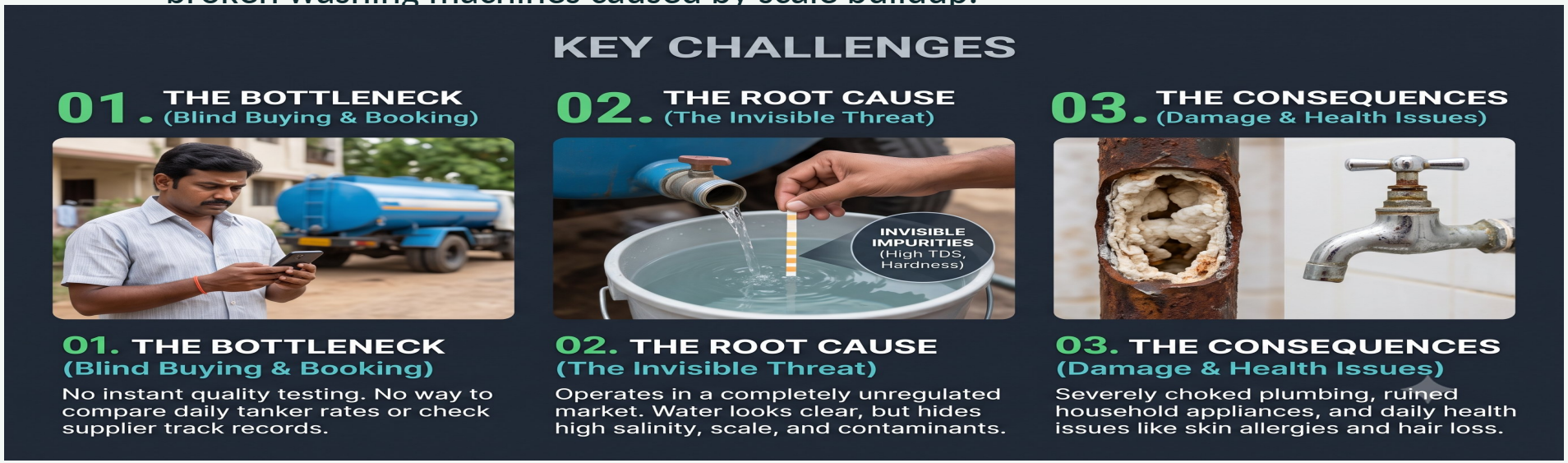
- The private water tanker industry is completely unorganized and operates in a "grey market" with zero government regulation or quality oversight.
- You can't "see" high water hardness, TDS, or dissolved chemicals with the naked eye, making it incredibly easy for suppliers to pass off bad groundwater as safe.

KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

Consequences of leaving them unsolved

- **Health:** Students and residents will continue suffering from severe hair fall, skin allergies, and long-term health impacts.
- **Financial:** Hostels and apartments will keep losing massive amounts of money—not just to surge pricing, but to constantly replacing choked plumbing, ruined water heaters, and broken washing machines caused by scale buildup.



EXISTING SOLUTIONS / APPROACHES

What is already being done, and where it falls short

Existing methods, systems or alternatives

- **Manual Booking:** Facility managers and residents rely on WhatsApp groups, building watchmen, or random phone numbers plastered on walls to find tankers.
- **Post-Delivery Treatment:** Households install expensive RO purifiers or individual bathroom water softeners to treat the water *after* they have already bought it.
- **Lab Testing:** Occasionally, large apartment complexes might send a water sample to a professional chemical testing lab.

Limitations of each

- **Manual Booking:** You have no idea if the driver is reliable, and prices fluctuate wildly based on your desperation and the driver's mood. There are no "Google Reviews" for water tankers.
- **Post-Delivery Treatment:** RO systems only fix *drinking* water. You are still showering and washing clothes in highly calcified water, which ruins plumbing and causes hair fall.
- **Lab Testing:** It is way too slow and expensive. A tanker delivers water today; a lab test takes 3 to 5 days. You can't wait a week to know if today's water is safe to use.

EXISTING SOLUTIONS / APPROACHES

What is already being done, and where it falls short

The gap they leave open (Key Challenges)

- **The "Point-of-Sale" Blindspot:** There is absolutely no tool for *instant*, *on-the-spot* quality verification right when the truck arrives.
- **No Community Power:** Consumers are completely isolated. Because there is no shared, public ledger of tanker prices and water quality, everyday buyers have zero negotiating power against unregulated







PROBLEM INSIGHTS & OPPORTUNITY AREAS

Resources required to solve the problem

Hardware Requirements

- **Zero custom hardware or IoT sensors are needed.** This keeps our prototype incredibly cheap and scalable.
- **Standard Smartphones:** Any basic Android or iOS device with a working camera to scan the water tests.
- **Off-the-shelf Chemical Test Strips:** Standard 5-in-1 aquarium or drinking water test strips (measuring pH, Hardness, TDS) which cost less than ₹10 per strip.

Software Requirements

- **Frontend Framework:** A mobile app or Progressive Web App (React Native or Flutter) to ensure an accessible, cross-platform user experience.
- **Computer Vision Pipeline:** Python and OpenCV (or a lightweight ML model) to color-calibrate and accurately read the chemical test strips under uneven lighting conditions.
- **Backend & Mapping:** A cloud database (like Firebase or AWS) to aggregate the crowdsourced ratings, integrated with a Maps API (like Google Maps or Mapbox) to display the live public ledger of tanker prices and quality.

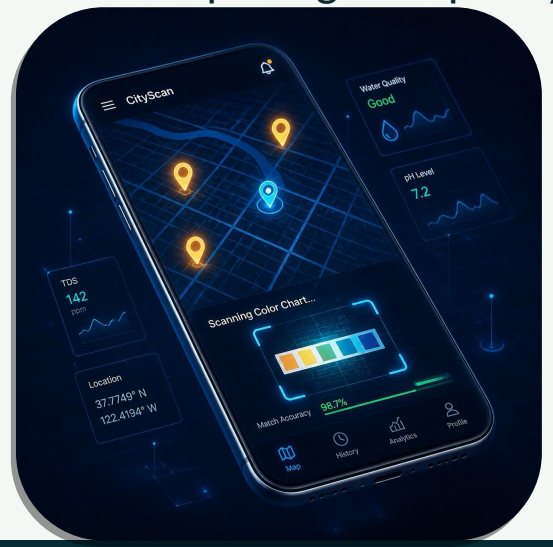
PROBLEM INSIGHTS & OPPORTUNITY AREAS

Resources required to solve the problem

Other resources (Including Human Resources)

- **Chemistry/Environmental Validation:** A brief consultation with a chemistry professor or environmental engineering student on campus to validate the accuracy of our test strip color ranges.
- **UI/UX Design:** Time dedicated to designing a completely foolproof, zero-friction interface so that non-technical users (like watchmen or hostel wardens) can easily scan and upload data.
- **Beta Testers:** Initial pilot testing with Amrita hostel facility managers or local Coimbatore Residential Welfare Associations (RWAs) to gather real-world pricing and quality data.







INITIAL DIRECTION & EXPECTED OUTCOME

What Is Your Expected Outcome?

What your team intends to do

- We intend to build a smartphone-based tool that uses computer vision to instantly read and analyze cheap chemical test strips right at the point of delivery.
- We will integrate this scanner with a crowdsourced dashboard that logs and displays live tanker prices, quality ratings, and supplier track records across the city.

What you want to improve

- We want to completely eliminate the "blind trust" forced upon consumers in the unregulated private water tanker market.
- We aim to improve consumer negotiating power by replacing hidden, volatile pricing and unknown water sources with complete, localized transparency.

INITIAL DIRECTION & EXPECTED OUTCOME

What Is Your Expected Outcome?

What successful improvement should you achieve

- **Instant Verification:** Empowering residents and facility managers to instantly reject highly calcified or contaminated water *before* it is dumped into their sumps.
- **Fair, Stabilized Pricing:** Eliminating arbitrary price gouging during dry spells by exposing scammers and driving community business toward ethical, highly-rated water suppliers.
- **Infrastructure & Health Protection:** Ultimately saving hostels and apartments thousands of rupees in ruined plumbing and preventing severe skin/hair issues caused by long-term hard water usage.

TEAM
Your Team Name

TEAM CONTRIBUTION

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Who will handle which part of the execution

Team Member	Role	Responsibility in Execution
Member 1 name	Role	What they will handle
Member 2 name	Role	What they will handle
Member 3 name	Role	What they will handle
Member 4 name	Role	What they will handle